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LETTER

Technology options and optimal pathways to a net-zero electricity system in Nigeria across different timelines

Michael O Dioha^{1,*} , Norbert Edomah² and Lily Odarno¹¹ Energy and Climate Innovation in Africa Program, Clean Air Task Force, Boston, MA, United States of America² School of Science and Technology, Pan-Atlantic University, Lagos, Nigeria

* Author to whom any correspondence should be addressed.

E-mail: mdioha@catf.us and michael@dioha.us**Keywords:** net-zero emissions, energy transition, CCS, nuclear, TIMES modelSupplementary material for this article is available [online](#)**Abstract**

Nigeria, one of Africa's largest economies, faces a pressing dual challenge: meeting its rapidly growing electricity demand while transitioning to a net-zero emissions electricity system. Using a stylized capacity expansion model, we explore pathways to a net-zero emissions electricity system by 2050, 2060, and 2070, examining the impacts of restricting firm low-carbon technologies such as nuclear power, gas with carbon capture and storage (gas-CCS), and concentrated solar power. Our results indicate that solar power, both utility-scale and off-grid, will be a pivotal component of Nigeria's future net-zero electricity mix. Across all net-zero scenarios, solar contributes between 37% and 55% of total electricity generation by 2050, and remains a major source of power in the 2060 and 2070 timeframes as well. Nuclear power plays a pivotal role in scenarios where it is allowed, accounting for 33%–41% of electricity generation by mid-century. In contrast, when nuclear is restricted, the system relies more heavily on gas-CCS and large-scale solar PV deployment. The cost analysis indicates that the transition to net-zero emissions electricity system requires average annual investments ranging from US\$7 billion to \$10 billion. The total system cost is consistently lower in net-zero emissions scenarios compared to business-as-usual, suggesting that transitioning to net-zero emissions, despite requiring higher initial investments, is more cost-effective in the long term for Nigeria. Earlier transitions do not necessarily lead to drastic higher system costs compared to delayed pathways. Sensitivity analyses reveal that $\pm 25\%$ variations in technology costs for solar, wind, and nuclear significantly influence the technology mix but do not alter the broader conclusion that firm low carbon technologies are vital to cost-effective transitions. We also assess the implications of high electricity demand growth consistent with Nigeria's Agenda 2050. The study provides actionable insights into the technology, investment, and policy trade-offs facing Nigeria and similar African economies in designing net-zero electricity systems.

1. Introduction

As electricity generation is responsible for more than 40% of carbon dioxide (CO₂) emissions associated with energy production (World Nuclear Association 2024), decarbonizing the electricity sector is a priority for most nations committed to reducing their environmental footprints. In Africa, the task of transforming the electricity sector is both an enormous challenge and a critical opportunity (Raji *et al* 2024). The continent is home to over one billion people, and its energy needs are growing rapidly due to significant population growth, urbanization, and industrialization (IEA 2022). Yet, Africa's electricity system is characterized by severe deficiencies in access, reliability, and affordability (IEA 2022, 2023a). These factors complicate the path to decarbonization, making the question of 'how' and 'when' Africa can achieve a net-zero emissions electricity system highly complex.

Taking a closer look at a true representation of Africa, Nigeria is the most populous and one of the largest economies in Africa. The electricity sector in Nigeria is a critical component of the country's overall energy system. The country's electricity generation capacity is significantly lower than demand, leading to frequent power outages and a heavy reliance on backup oil generators, which are both 'expensive' and carbon-intensive (Nana and Dioha 2024). Currently, over 25% of Nigerians lack access to electricity (IEA 2023a). Despite being endowed with abundant energy resources, including natural gas, solar, and hydroelectric potential, Nigeria has struggled to develop a reliable and sufficient power supply system (Edomah 2016). A detailed review of the Nigerian electricity sector is given in (Babatunde *et al* 2023). At the same time, the nation faces the urgent task of reducing CO₂ emissions in alignment with global climate goals (Dioha 2020).

In response to these challenges, Nigeria, and indeed many other countries in Africa has committed to increasing its electricity generation capacity while simultaneously reducing the carbon intensity of its electricity mix. Nigeria's climate policy commits the nation to achieving net-zero emissions economy between 2050–2070. Moreover, the country's energy transition plan (ETP) which is framed by its commitments under the Paris Agreement, as well as its nationally determined contributions, aims to achieve net-zero emissions energy system by 2060 (Dioha 2022, FGN 2022). Achieving a net-zero emissions electricity system will require a strategic overhaul of the existing energy system. Central to this challenge is the timeline for decarbonization, which is essential for decarbonizing the electricity sector. The uncertainties surrounding the timeline for achieving net-zero emissions in Nigeria's electricity sector can be traced to several factors, including economic, infrastructural, technical, and policy-related barriers (Dioha and Kumar 2020).

On the economic front, the high initial costs of certain low carbon technologies such as concentrated solar power (CSP) is a significant barrier to their deployment in Nigeria. The Nigerian government has limited financial resources given the country's fiscal constraints and competing priorities such as health-care and education and struggle to attract investment for expensive low carbon energy infrastructure projects (Dioha and Kumar 2020, Dioha 2022). Also, the cost of capital is relatively high in Nigeria due to perceived risks of investments in the country (Ibrahim *et al* 2021). This reduces the attractiveness of long-term low carbon energy technology projects, which require substantial upfront investment and have extended payback periods. In terms of socio-political challenges, key low carbon technologies like nuclear power projects face significant political and social opposition due to concerns about safety, environmental impact, and the potential for accidents (Ewim *et al* 2022). Also, in terms of technical capacity, technologies such as carbon capture & storage (CCS) is complex and requires specialized infrastructure and expertise, which is lacking in Nigeria (Galadima and Garba 2008, Nwali *et al* 2024). These challenges put a spotlight on the feasibility of deploying these firm low carbon technologies (nuclear, CCS, and CSP) and when can Nigeria achieve a net-zero emissions electricity supply system.

Energy system models are computational tools used to explore transformation pathways for energy systems (Dioha 2017, Tesfamichael and Fuchs 2024). Over the years, a plethora of studies have employed these models to study Nigeria's electricity system long-term transformation. Electricity system transformation pathways studies for Nigeria has been focused on the power sector as a stand-alone work (Oyewo *et al* 2018, Roche *et al* 2020, Tambari *et al* 2020, Daggash and Mac Dowell 2021, Kim *et al* 2021, Bugaje *et al* 2022, Nana and Dioha 2024) or part of an economy-wide energy system analysis study (Emodi *et al* 2017, Dioha *et al* 2019, Dioha and Kumar 2020, IRENA 2023). The methodological approaches of these studies differ, ranging from cost optimization capacity expansion modeling such as (Tambari *et al* 2020) to simulation techniques such as (Roche *et al* 2020), as well as other analytical methods like scenario analysis (Emodi *et al* 2017, Daggash and Mac Dowell 2021). The key findings from these studies indicate that fuel and technology switching, alongside capital investments in renewable energy and critical low-carbon power sector technologies—such as battery storage and nuclear power will be the primary drivers of decarbonization efforts in Nigeria's electricity generation system. As a result, the power sector emerges as a foundational element for achieving net-zero emissions in the country. However, none of these studies have critically examined the techno-economic impact of achieving net-zero emissions across different timelines and their implications for Nigeria's electricity system decarbonization trajectory. Moreover, several pathways to decarbonization exist, but their feasibility, costs, and impacts differ significantly depending on the timelines set for achieving net-zero emissions.

Here, we explore the optimal mix of technologies and costs of achieving a net-zero emissions electricity system in Nigeria across different timelines, examining scenarios for near-term (2050), mid-term (2060), and long-term decarbonization (2070). In that regard, we develop a stylized capacity expansion model for Nigeria's electricity generation system using the TIMES (The Integrated MARKAL-EFOM System) modeling framework (Loulou *et al* 2016). We explore the least-cost technology configuration for Nigeria's electricity system under a business-as-usual (BAU) scenario without net-zero target. Next, we

explore net-zero emissions scenarios by 2050, 2060, and 2070 distinctively considering the availability of all modeled technologies including key non-conventional low carbon technologies such as nuclear, CSP and CCS. We also explore variants of the 2050, 2060 and 2070 net-zero scenarios considering restrictions on the deployment of nuclear, CSP and CCS. Results are assessed in terms of electricity generation capacity and energy system costs. Finally, we discuss the policy implications of our findings.

This study is stylized as it does not aim to forecast the evolution of Nigeria's electricity system but aims to provide indicative macro techno-economic implications of the system under a net-zero framework. Thus, the crux of this analysis is to conduct a 'what if' scenario evaluation to assess how differing decarbonization timelines and limitations in the deployment of non-conventional firm low carbon technologies could influence Nigeria's ability to achieve net-zero emissions electricity system. This analysis aims to provide additional insights for energy policy discussions, exploring alternative technological pathways that could complement existing energy sector planning strategies. Also, this study significantly advances academic literature by offering critical insights into low-carbon transitions in developing countries, with a focus on Nigeria. Its findings contribute to the global discourse on climate goals and offer a valuable framework for future research.

2. Methods

2.1. The TIMES modeling framework

The TIMES model generator was developed by the Energy Technology Systems Analysis Program (ETSAP), a Technology Collaboration Program under the International Energy Agency (IEA). ETSAP is a global network that leverages long-term energy scenarios to perform comprehensive analyses of energy and environmental systems. The TIMES model generator integrates two distinct yet complementary approaches to energy modeling: a technical engineering perspective and an economic perspective. TIMES is a technology-rich, bottom-up model generator that employs linear programming to develop an optimized, least-cost energy system, tailored to a range of user-defined constraints, over medium- to long-term timeframes (Loulou *et al* 2016).

The objective function of the TIMES model (equation (1)) seeks to minimize the total discounted cost of the energy system, while being faithful to a range of technical and environmental constraints specified within the model. Fundamentally, this objective function equates to maximizing the net total surplus, which includes both consumer and producer surpluses.

$$NPV = \sum_{r=1}^R \sum_{y \in \text{YEARS}} (1 + d_{r,y})^{\text{REFYR}-y} \times \text{ANNCOST}(r,y) \quad (1)$$

where NPV represents the objective function of TIMES (i.e. the net present value of the total cost of all regions); R the number of regions in the study area; YEARS the set of years in the model for which there are costs; $d_{r,y}$ the discount rate; REFYR the reference year for discounting; and ANNCOST(r,y) the total annual cost in region r and year y .

Like its predecessor MARKAL (MARKet ALlocation), TIMES can represent the entire energy system, from resource supply through fuel processing, infrastructure representation, and conversion to secondary energy carriers, to end-use technologies and energy service demands across various sectors. This makes TIMES an ideal framework for developing an integrated model of the entire energy system. Additionally, TIMES is well-suited for detailed modeling of specific subsectors, such as the electricity system. As a perfect foresight model, TIMES assumes that participants in the system possess complete inter-temporal knowledge of future demand, technology, policy, and other factors. Consequently, TIMES identifies the cost-optimal development of energy system across the economy, based on a given set of input assumptions, and generates detailed outputs on the evolution of the energy system. The TIMES modeling framework has proven to be a valuable tool for various applications and is increasingly used for national, regional, and global analyses, such as those conducted by the IEA and IPCC (ETSAP 2023). A key strength of the TIMES optimization process is its systematic approach to uncertainty, which is addressed through 'what-if' analyses designed to quantify scenarios in the transition between technology categories and energy pathways. Detailed model equations and documentation of TIMES is available on ETSAP website (<https://iea-etsap.org/index.php/documentation>) and in Loulou *et al* (2016). TIMES code is publicly available at https://github.com/etsap-TIMES/TIMES_model.

2.2. Nigeria TIMES electricity capacity expansion model

The Nigerian electricity system is modeled in a stylized manner using the TIMES modeling framework. The developed reference energy system (RES) for Nigeria's electricity generation system is illustrated in

figure 1. This RES encompasses the full spectrum of energy activities, extending from primary energy sources to the final end-use electricity demand. Among the various input data, electricity demand is particularly significant, as it is regarded as the primary driving factor for the model. A recent comprehensive bottom-up analysis, Nigeria's renewable energy roadmap, conducted by the International Renewable Energy Agency (IRENA) in collaboration with the Energy Commission of Nigeria (ECN), evaluated the baseline future electricity demand in Nigeria. The study indicated that Nigeria's total electricity demand, which was 113 TWh in 2020, could rise to 520 TWh by 2050 (IRENA 2023). This projection covers various sectors, including residential, industrial, commercial, transport, and agriculture. In this study, just as in Nana and Dioha (2024), we linearly extrapolate IRENA's electricity demand projection through to 2070, estimating future demand to reach 876 TWh (figure 2). This corresponds to a compound annual growth rate (CAGR) of approximately 4.2%. An analysis of Nigeria hourly electricity demand data (Brinkerink and Deane 2020, Brinkerink *et al* 2021) was conducted to derive the demand profile.

The model is configured for the base Nigerian electricity generation system and employs a system-wide discount rate of 10% (Tambari *et al* 2020, Daggash and Mac Dowell 2021, IRENA 2023). The model includes both non-renewable (e.g. natural gas and nuclear) and renewable (e.g. solar, wind, hydro, and biomass) energy carriers, which serve as inputs for electricity generation technologies. Hourly solar and wind capacity factors for Nigeria are derived from www.renewables.ninja/.

The modeling period spans from 2020 to 2070 (with five years intervals). When considering time representation, it is crucial to balance the increased value of adding more timeslices with factors such as model solution time, the complexity introduced, and the insights gained. Here, inter-annual timeslices are represented at seasonal, daily, and hourly levels. Nigeria experiences two primary seasons: the dry season, generally occurring from November to March, and the rainy season, typically spanning from April to October (Ibebuchi and Abu 2023). We consider these two seasons (dry and wet) in our study and then use *K*-means clustering to select three representative days in each season considering their 24 h profiles (details of *K*-means cluster analysis is presented in the Supplementary information). Thus, the model has 144 timeslices (table S2).

The modeling framework requires detailed characteristics of all conversion technologies, which are essential for converting primary energy into electrical energy for end-use services. The Nigerian capacity expansion model integrates a representative set of conversion technologies, including large and small hydro, natural gas, biomass, nuclear, solar, wind, battery storage, and transmission and distribution technologies. Table 1 provides comprehensive details on the techno-economic parameters of various conversion technologies, including efficiency, capital cost, operation & maintenance (O&M) cost etc. Recognizing the significant role of off-grid systems within Nigeria's power sector, the model also incorporates decentralized technologies, such as captive diesel and gasoline (oil) generators, and off-grid solar PV systems. Tables 2 presents the techno-economic parameters of the transmission and distribution technologies. Table 3 presents the fuel prices applied in this study.

In the TIMES model, constraints are introduced on technology adoption to more accurately reflect real-world conditions, recognizing that factors beyond cost can influence the deployment of technologies (Dioha *et al* 2023). Accordingly, existing technologies are retained in the system until their retirement, while the starting year for investment in new technologies is specified in table 1. The delays in starting years of some technologies also mirror Nigeria's own plans. For instance, the Nigerian ETP has no meaningful utility solar PV in the mix until 2030 (FGN 2022). Electricity technologies like nuclear, gas-CCS, and CSP are also absent in Nigeria presently (ECN 2022). Their future start years in our model is due to current institutional hurdles facing the Nigerian energy sector. A recent study highlights that Nigeria's energy sector suffers from 'overlapping institutional and regulatory regimes' and inadequate funding (Gungah *et al* 2019). Until these governance gaps are addressed, ambitious roll-out of nuclear or CSP or gas-CCS is unlikely. In sum, deferring CSP until 2035 and batteries until 2030 aligns with the reality that Nigeria must first develop robust policy frameworks for them backed by adequate financing. In addition, to improve the realism of the modeled scenarios, upper limits are also imposed on the installed capacities of various technologies by 2050 and extrapolated to 2070. These assumptions are derived from the ECN's modeling study—Nigeria Energy Calculator 2050 (ECN 2015, Dioha *et al* 2019). The upper limits are shown in table S1.

2.3. Scenarios design

We build two core scenarios: the BAU scenario and the net-zero emissions scenarios. The BAU scenario is based on the planned energy scenario of the ECN/IRENA study of Nigeria's renewable energy

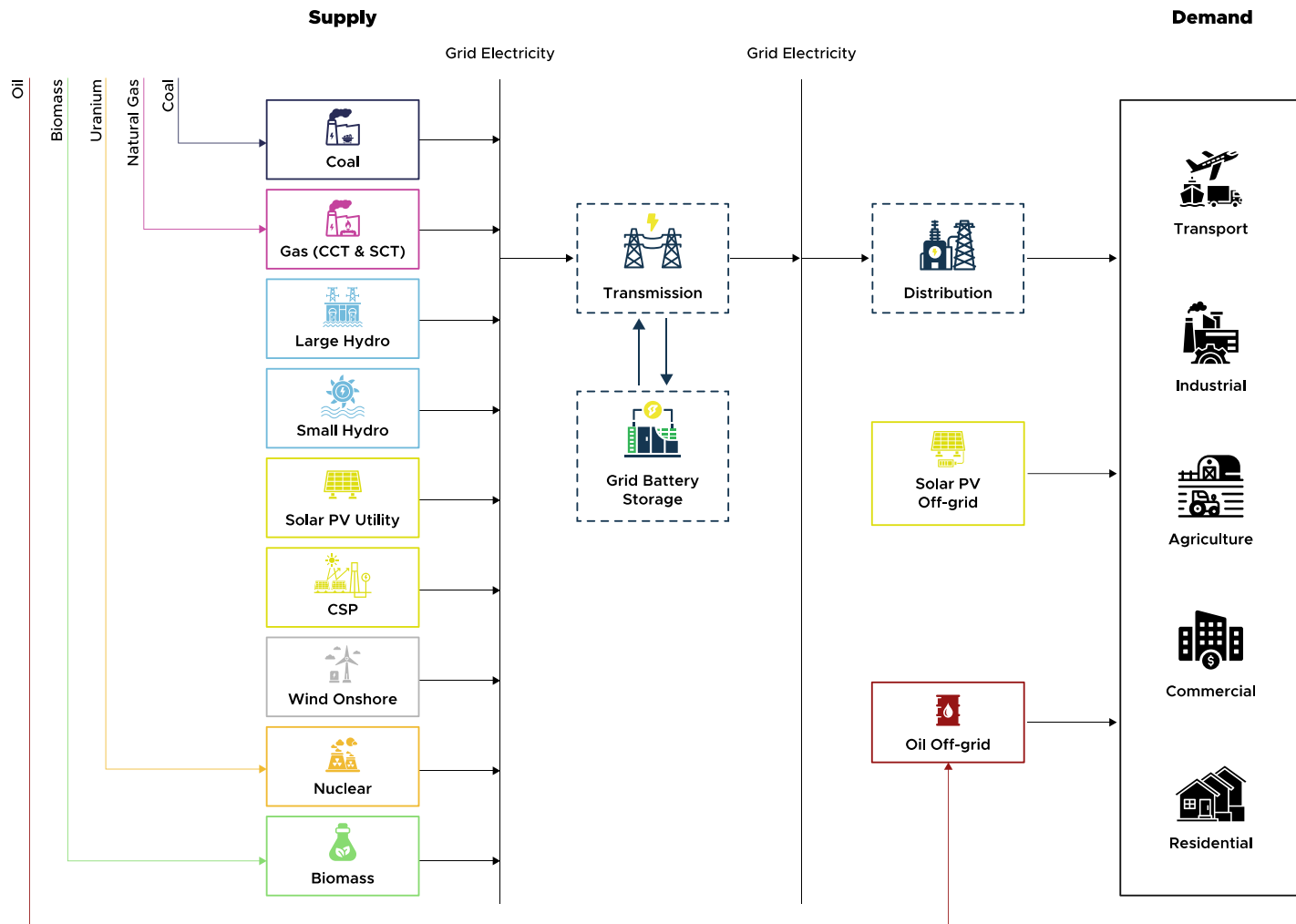


Figure 1. Simplified structure of the Nigeria TIMES electricity generation model.

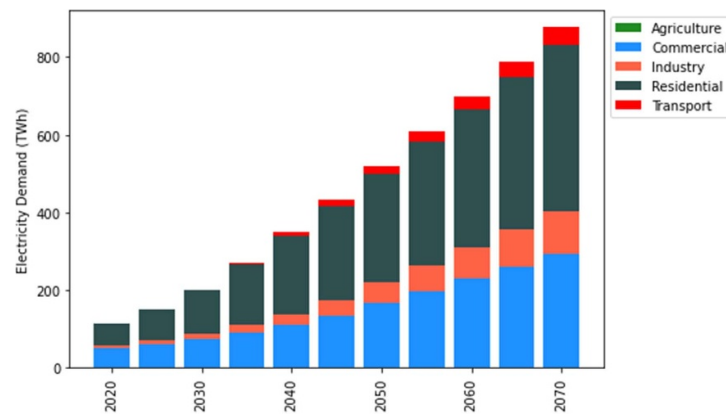


Figure 2. Nigeria electricity demand projection.

roadmap which was adapted for this study (IRENA 2023). This scenario is based on policies and measures that were implemented prior to 2020. The existing policies encompass ongoing energy/climate initiatives, as well as those outlined in Nigeria's energy master plan (ECN 2022). This scenario contains no net-zero emissions target.

Nigeria's Climate Change Act outlines a national vision for attaining a net-zero emissions energy system within the timeframe of 2050–2070 (Climate Action Tracker 2022). However, the adoption of a net-zero emissions target in Nigeria presents both significant opportunities and challenges. On one hand, transitioning to a low-carbon electricity system can deliver multiple benefits, including reduced air pollution and improved energy security (Dioha *et al* 2019, Tambari *et al* 2020). On the other hand, the transition requires careful consideration of the country's developmental priorities, particularly in relation to poverty reduction, job creation, and industrialization (Dioha 2022). Nigeria's energy policy must therefore strike a balance between mitigating emissions and promoting economic and social development which could influence the timeline of achieving a net-zero emissions electricity system. Considering Nigeria's climate act and given the urgency of addressing socio-economic challenges, three potential timelines scenarios for achieving net-zero emissions (CO₂ in this case) in the electricity sector have been analyzed. These timelines have been framed here as immediate action (achieving net-zero by 2050), moderate action (achieving net-zero by 2060), and delayed action (achieving net-zero by 2070).

Achieving net-zero emissions electricity could hinge on the large-scale deployment of key non-conventional firm low carbon technologies such as nuclear, CSP and gas-CCS. However, the deployment of these technologies in Nigeria faces significant challenges. For instance, Nigeria currently has no CCS and nuclear power in place (Ewim *et al* 2022, FGN 2022). Consequently, for each net-zero timeline scenario, we analyzed other variants of the scenario considering restrictions³ in the deployment of firm low carbon generators (nuclear, CSP, and natural gas equipped with CCS). Table 4 summarizes the scenarios design.

2.4. Sensitivity analysis

In addition to our core scenarios shown in table 4, we conducted a series of sensitivity analyses focusing on two dimensions: (1) variation in capital investment costs, and (2) future electricity demand projections influenced by national development ambition.

1. Technology investment cost sensitivities: Uncertainty about future technological costs remains a significant factor in long-term energy system modeling, particularly in Africa where cost trajectories can vary due to market maturity, local supply chains, and high cost of capital (Dato *et al* 2024). Nuclear power is highly capital-intensive, with large upfront costs incurred before any revenue is generated (World Nuclear Association 2025). Also, even though renewable energy costs have declined substantially in recent years, future cost trajectories remain uncertain. Factors such as supply chain disruptions and tariffs may impact their investment costs. To capture these uncertainties, we performed sensitivity analyses on the investment costs of nuclear power and solar (utility and off-grid) and onshore wind technologies. We implemented a $\pm 25\%$ variation in investment costs for nuclear power and solar (utility and off-grid)

³ By 'restriction,' we refer to the explicit prohibition of the technology within the modeled scenario, thereby excluding it from consideration or implementation in the scenario analysis.

Table 1. Assumptions on techno-economic parameters of electricity generation technologies and storage.

Technology	Investment cost (\$/kW)			Fixed O&M cost (\$/kW-yr.)	Average capacity factor	Technical lifetime (Years)	Thermal efficiency	Start year for new capacity
	2020	2050	2070					
CCGT ^a	1221	1072	1022	16.0	0.60 ^d	35	0.47	2020
SCGT ^a	799	694	694	12.0	0.60 ^d	35	0.32	2020
CCGT + CCS ^b	2845	2000	1900	28.9	0.60 ^d	35	0.47	2035
Coal	2500	2500	2500	34.0	0.70 ^d	50	0.34	2030
Large hydro	3000	3000	3000	21.9	0.38	50	—	2020
Small hydro	3319	3279	3252	25.8	0.60	45	—	2020
Solar PV—utility ^c	1378	723	642	11.0	—	25	—	2025
Concentrated solar power	5797	4294	3660	178.0	0.45	25	—	2035
Solar PV—offgrid	4320	2091	2091	86.0	0.34	25	—	2020
Wind-onshore ^c	1730	1126	723	60.0	—	25	—	2025
Nuclear	6137	6137	6137	91.3	0.85	60	0.33	2035
Biomass	2534	2291	2129	64.0	0.50	30	0.31	2025
Oil	750	750	750	23.0	0.30	8	0.16	2020
Battery storage—4 h	1287	307	273	26.0	—	15	0.85 ^e	2030

Data Source: (Cole and Frazier 2019, Dioha and Kumar 2020, Allington *et al* 2023).

^a Combined cycle gas turbines (CCGT) and single cycle gas turbines (SCGT). It is assumed that new investments will favor CCGTs over SCGTs since CCGTs are relatively more efficient.

^b The current capture rate of carbon capture and storage (CCS) for power generation is typically between 85% and 95%, depending on the technology and plant efficiency. However, this stylized study assumes a 100% capture rate. In Nigeria, research on CO₂ capture, transport, and storage is still in its early stages and there is no reliable estimate of the country's cumulative CO₂ storage potential, specifically for the power sector. The associated cost for CO₂ transport and storage is taken as US\$ 15 per tonne (Smith 2021, Das *et al* 2023).

^c Solar PV utility and wind-onshore grid integration costs are taken as 5% and 11% of the investment costs respectively (IEA-ETSAP 2015).

^d Capacity factor assumed to improve in the future.

^e Round-trip efficiency.

Table 2. Techno-economic assumptions of transmission and distribution technologies.

Technology	Capital cost (\$/kW)	Variable cost (\$/GJ)	Operational life (years)	Efficiency (2020)	Efficiency (2050)	Efficiency (2070)
electricity transmission	365	0.0001	50	90%	95%	95%
Electricity distribution	2502	0.0001	70	91%	94%	94%

Data Source: (Pappis *et al* 2019, Allington *et al* 2023).

Table 3. Projection of fuel prices (US\$/GJ).

Fuel	2020	2050	2070
Natural gas	7.1	9.9	9.9
Biomass (Agricultural residues)	1.6	1.6	1.6
Oil	14.7	23.6	30
Uranium	0.28	0.28	0.28
Coal	1.4	1.6	1.7

Data Source: (Dioha and Kumar 2020, NEA 2020, Allington *et al* 2023).

and onshore wind technologies under the NZ50_All, NZ60_All, and NZ70_All scenarios. We aimed to assess how cost fluctuations may alter technology deployment patterns, system-wide generation mixes, and overall investment needs.

2. Demand-driven sensitivities: A second dimension of sensitivity analysis focuses on the evolution of electricity demand, which is a crucial determinant of system capacity requirements. Nigeria's 'Nigeria Agenda 2050'—a long-term strategic development plan which sets out a vision for transforming the country into a dynamic, industrialized, and knowledge-driven economy (FGN 2023). A central target of this agenda is to achieve a per capita income of US\$33 328 by 2050, which would place Nigeria in the high-income country category based on current World Bank country classifications (The World Bank 2025). Historically, there is a strong positive correlation between income levels and electricity consumption. High-income countries typically exhibit significantly higher electricity usage, driven by industrial activity and lifestyle changes. Nigeria's per capita electricity consumption is approximately 529 kWh/capita (including off-grid consumption) (IRENA 2023), which is markedly lower than the median of high-income countries [6720 kWh/capita (Moss *et al* 2021)]. To reflect the possible implications of this development trajectory, we assume Nigeria achieves the median per capita electricity consumption of high-income countries by 2050 and maintains this level through to 2070. This scenario is consistent with the ambitions articulated in the Nigeria Agenda 2050. This demand sensitivity is also applied to the NZ50_All, NZ60_All, and NZ70_All scenarios.

3. Results

The scenario results are analyzed based on changes in installed electricity generation capacity, generation mix, and energy system costs. For the sake of analytical tractability and conciseness, the main article focuses on the results of BAU scenario and the all-technologies net-zero scenarios for 2050, 2060, and 2070. Results for the corresponding scenario variants and sensitivity analysis are provided in the supplementary material.

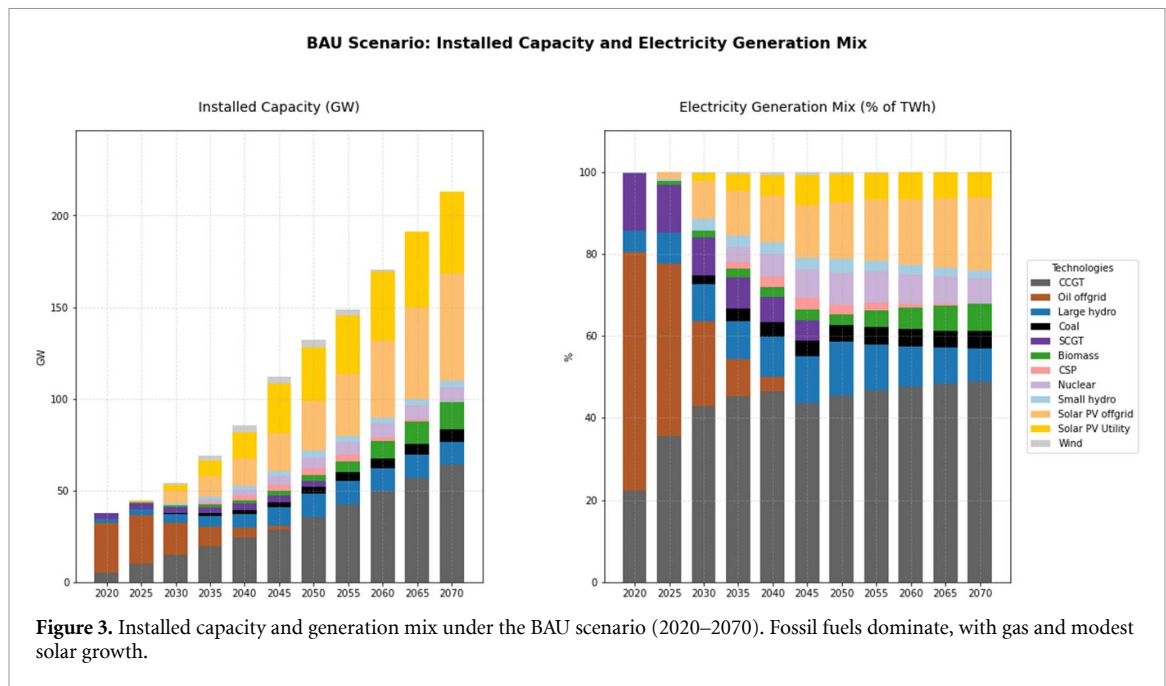
3.1. Installed electricity capacity and electricity generation mix

3.1.1. BAU

Figure 3 shows Nigeria's electricity sector is poised for significant growth and transformation in the coming decades. In the BAU (figure 3), without any net-zero emissions target, total installed capacity grows over 5-fold to meet rising electricity demand, reaching 213 GW by 2070 (up from about 38 GW in 2020). In 2020, the total installed capacity of 38 GW encompassed off-grid and grid connected systems. The off-grid system is primarily dominated by off-grid gasoline and diesel (oil) generators. The oil-fired generators contributed 72% to the total installed capacity in 2020. The expansion under BAU is characterized by heavy reliance on fossil fuels, especially natural gas (unabated), complemented by moderate growth in renewables and other sources. Combined-cycle turbines (CCGT) capacity increases from ~5 GW in 2020 to 64 GW by 2070. There is no CCS applied to gas in BAU, so all gas capacity is emitting. In 2020 a notable portion (27 GW) of capacity is diesel and gasoline generators (oil off-grid), but this is rapidly diminishing over time and is completely phased out by 2045. Off-grid oil is replaced largely by solar PV off-grid. Beyond the least-cost approach reflected in the model, these findings align

Table 4. Summary of scenarios.

Scenarios	Scenario variant	Description
Business-as-usual (BAU)		Follows an adapted ECN/IRENA planned energy scenario for Nigeria (IRENA 2023). No net-zero CO ₂ emissions electricity system target.
2050 net-zero all (NZ50_All)	2050 net-zero restricted nuclear (NZ50_restr_nuc)	Nigeria achieves net zero CO ₂ emissions electricity system by 2050 without any restrictions on modeled technologies.
	2050 net-zero restricted nuclear + CCS (NZ50_restr_nuc + CCS)	Nigeria achieves net zero CO ₂ emissions electricity system by 2050 with restricted investment in nuclear power.
	2050 net-zero restricted nuclear + CSP (NZ50_restr_nuc + CSP)	Nigeria achieves net zero CO ₂ emissions electricity system by 2050 with restricted investment in nuclear power and gas/wCCS
	2050 net-zero restricted CCS + CSP (NZ50_restr_ccs + CSP)	Nigeria achieves net zero CO ₂ emissions electricity system by 2050 with restricted investment in nuclear power and CSP
	2050 net-zero restricted CCS + CSP (NZ50_restr_ccs + CSP)	Nigeria achieves net zero CO ₂ emissions electricity system by 2050 with restricted investment in gas/wCCS and CSP
2060 net-zero all (NZ60_All)	2060 net-zero restricted nuclear (NZ60_restr_nuc)	Nigeria achieves net zero CO ₂ emissions electricity system by 2060 without any restrictions on modeled technologies.
	2060 net-zero restricted nuclear + CCS (NZ60_restr_nuc + CCS)	Nigeria achieves net zero CO ₂ emissions electricity system by 2060 with restricted investment in nuclear power.
	2060 net-zero restricted nuclear + CSP (NZ60_restr_nuc + CSP)	Nigeria achieves net zero CO ₂ emissions electricity system by 2060 with restricted investment in nuclear power and gas/wCCS
	2060 net-zero restricted CCS + CSP (NZ60_restr_ccs + CSP)	Nigeria achieves net zero CO ₂ emissions electricity system by 2060 with restricted investment in nuclear power and CSP
	2060 net-zero restricted CCS + CSP (NZ60_restr_ccs + CSP)	Nigeria achieves net zero CO ₂ emissions electricity system by 2060 with restricted investment in gas/wCCS and CSP
2070 net-zero all (NZ70_All)	2070 net-zero restricted nuclear (NZ70_restr_nuc)	Nigeria achieves net zero CO ₂ emissions electricity system by 2070 without any restrictions on modeled technologies.
	2070 net-zero restricted nuclear + CCS (NZ70_restr_nuc + CCS)	Nigeria achieves net zero CO ₂ emissions electricity system by 2070 with restricted investment in nuclear power.
	2070 net-zero restricted nuclear + CSP (NZ70_restr_nuc + CSP)	Nigeria achieves net zero CO ₂ emissions electricity system by 2070 with restricted investment in nuclear power and gas/wCCS
	2070 net-zero restricted CCS + CSP (NZ70_restr_ccs + CSP)	Nigeria achieves net zero CO ₂ emissions electricity system by 2070 with restricted investment in nuclear power and CSP
	2070 net-zero restricted CCS + CSP (NZ70_restr_ccs + CSP)	Nigeria achieves net zero CO ₂ emissions electricity system by 2070 with restricted investment in gas/wCCS and CSP

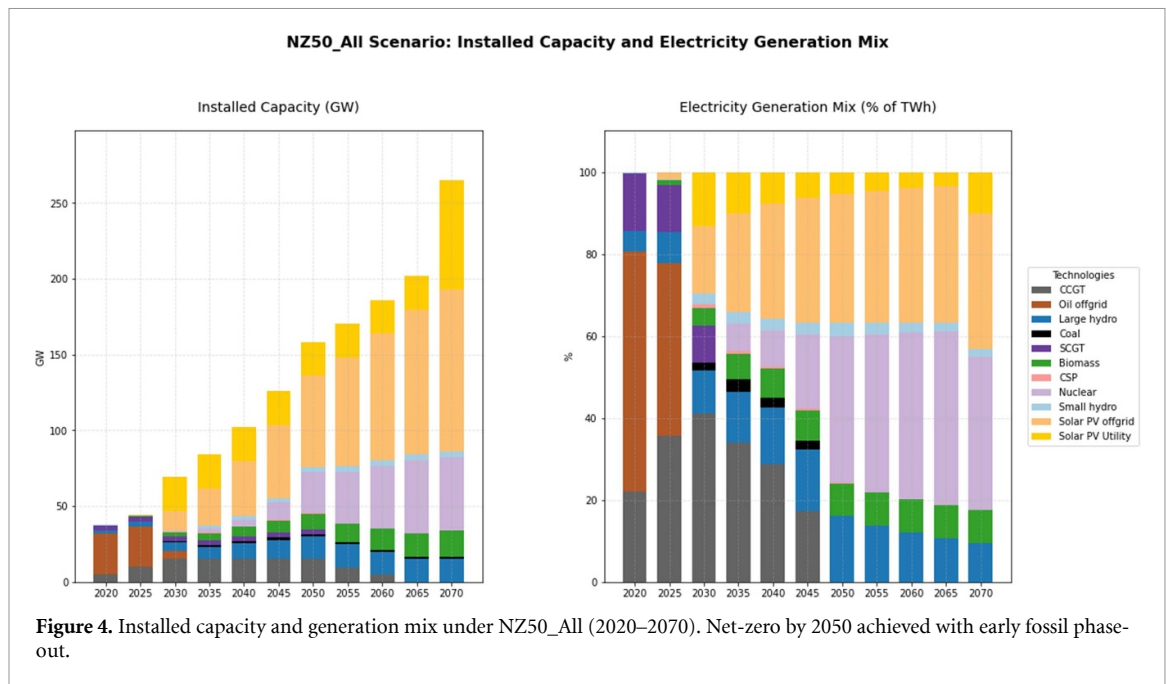


with recent trends in the Nigerian electricity sector. Following the recent removal of fuel subsidies (Musa and Abubakar 2025), the operational costs of backup oil-fired generators have significantly increased, rendering them no longer cost-effective. As a result, many households and small businesses are transitioning to off-grid solar PV systems as a more viable backup solution for the unreliable grid supply. The BAU shows growing renewables, but at a more modest pace than in ‘restricted’ scenarios. Both utility-scale and off-grid solar expand, especially after 2030. Utility-scale PV grows from virtually 0 in 2020 to 29 GW by 2050.

BAU (figure 3) shows Nigeria could leverage its hydro resources but does not fully maximize them by mid-century. Large hydro capacity rises from 1.9 GW in 2020 to 12.5 GW by 2050 and remains at 12.5 GW thereafter (2060–2070). Wind (onshore wind) plays only a minor role. BAU sees a small addition of wind. In fact, the wind capacity actually declines to 0 by 2070 (as those initial turbines retire and no new ones are built). Biomass power plants start appearing in the 2030s and reach 3.4 GW by 2050. Biomass capacity continues to grow post-2050 in BAU, up to 15 GW by 2070, representing biomass-fueled generation that provides dispatchable renewable power. A small coal capacity is introduced in BAU and reaches only 6.8 GW by 2070 (with 3.8 GW by 2050). Coal remains a minor player due to Nigeria’s preference for gas as also shown in its ETP objectives (FGN 2022). Interestingly, BAU incorporates a nascent nuclear program. Starting around the mid-2030s, nuclear capacity grows to 6.3 GW by 2050 and about 7.4 GW by 2060, and upto ~8 GW through 2070. This suggests that even without a net-zero mandate, Nigeria might pursue nuclear for least-cost energy security reasons.

3.1.2. Net-zero by 2050

Figure 4 shows results of the NZ50_All scenario (all modeled technologies available). In the NZ50_All, Nigeria reaches net-zero CO₂ in the electricity system by 2050 through a combination of early fossil-fuel phase-out and massive build-out of low-carbon capacity. By 2050, total installed capacity is about 158 GW (20% higher than BAU’s in 2050), and it grows to roughly 265 GW by 2070. In NZ50_All, CCGT still reaches 15 GW in 2030 (similar to BAU up to that point), but unlike BAU, no further gas expansion occurs after 2030. In the NZ50_All scenario, fossil fuel-based electricity is significantly reduced, with CCGT being entirely phased out, leaving the installed 15 GW in 2050 as stranded capacity. As restrictions are introduced in other NZ50 scenarios (see results in supplementary file), the stranded capacity increases up to 19.4 GW (NZ50_Restr_Nuc + CSP). With fossil power plants constrained, nuclear power becomes a cornerstone of firm low-carbon generation in NZ50_All. Starting from zero, nuclear capacity ramps up dramatically in the 2040s and reaches 27.2 GW by 2050. All available hydro-power is exploited by 2050. Similarly, biomass power is accelerated to ~10 GW in 2050 (versus 3.4 GW in BAU in 2050).

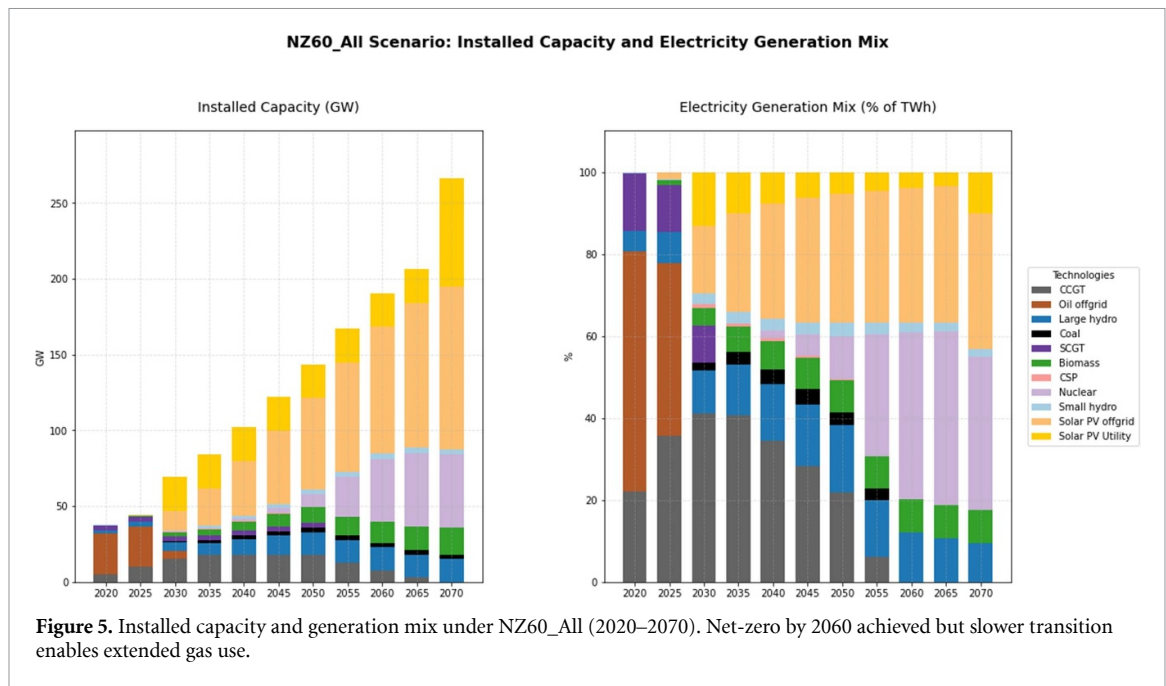


Solar PV plays a crucial role but interestingly, the distribution between utility-scale and off-grid PV differs from BAU due to the substantial presence of nuclear. Off-Grid Solar PV sees a much larger adoption by 2050 than in BAU—about 60 GW off-grid PV in 2050 (more than double the BAU’s 27 GW). This reflects an aggressive push to replace remaining off-grid diesel and gasoline gensets. By 2070, off-grid PV rises further to 107 GW in this scenario. Utility-scale solar PV, however, is initially ‘lower’ in NZ50_All than in BAU because nuclear is doing much of the heavy lifting for grid supply by 2050. Utility PV capacity in 2050 is only about 22.4 GW (compared to 29 GW in BAU 2050). With abundant nuclear, the model does not need to push utility solar as hard by 2050. Nonetheless, utility PV expands later to meet growing demand and make up for any remaining fossil generation. So, while nuclear’s presence delayed some PV investment early on, in the long run a substantial solar build-out still occurs. The total solar (utility + off-grid) by 2070 is 179 GW, making solar PV (especially off-grid) one of the largest contributors to capacity. Additionally, in NZ50_All, wind is effectively absent due to its limited cost-effectiveness, which is largely attributable to Nigeria’s relatively poor wind regime (Dioha *et al* 2016). This finding is in line with the results of (Duan *et al* 2022) that show that nuclear power can play a valuable role in achieving zero-emission electricity systems, particularly in regions with limited wind energy potential.

CSP is also almost non-existent in this scenario—only a token 0.5 GW CSP plant appears around 2030–2050 and is then phased out. Since nuclear (and to an extent hydro/biomass) can provide firm power, expensive CSP is not needed to achieve net-zero, and the model opts not to deploy it at scale. No gas with CCS is built either in NZ50_All (since nuclear is cheaper or preferred to handle the load, and any remaining gas was simply eliminated rather than cleaned). Coal and oil are essentially 0 by 2050 in this scenario. This represents a complete turnaround from the unabated gas-heavy BAU mix. It is worth noting that although net-zero is reached by 2050, electricity demand keeps growing, so from 2050–2070 the NZ50_All scenario continues to add capacity—all of it in low-carbon forms. The results of Net-Zero 2050 scenarios with technology restrictions are presented in the supplementary file (section 1.1).

3.1.3. Net-zero by 2060

Figure 5 presents the results of the NZ60_All scenario while the results of Net-Zero 2060 scenarios with technology restrictions are presented in the supplementary file (section 1.2). Generally, the Net-Zero 2060 scenarios give a decade more breathing room than 2050. This allows a slower initial deployment of low-carbon capacity and a later phase-out of fossils, though by 2060 the power sector must still be at net-zero. The additional time changes the trajectory of capacity build-out, but interestingly, by around 2070 the total installed capacity ends up similar to the 2050 case (because ultimately to meet growing demand with zero emissions, similar infrastructure is needed—just deployed later). In the NZ60_All (figure 5); key differences from the 2050 scenario include that with net-zero not required until 2060, unabated gas (CCGT) can operate longer. CCGT capacity still grows to ~15 GW by 2030 (same as other



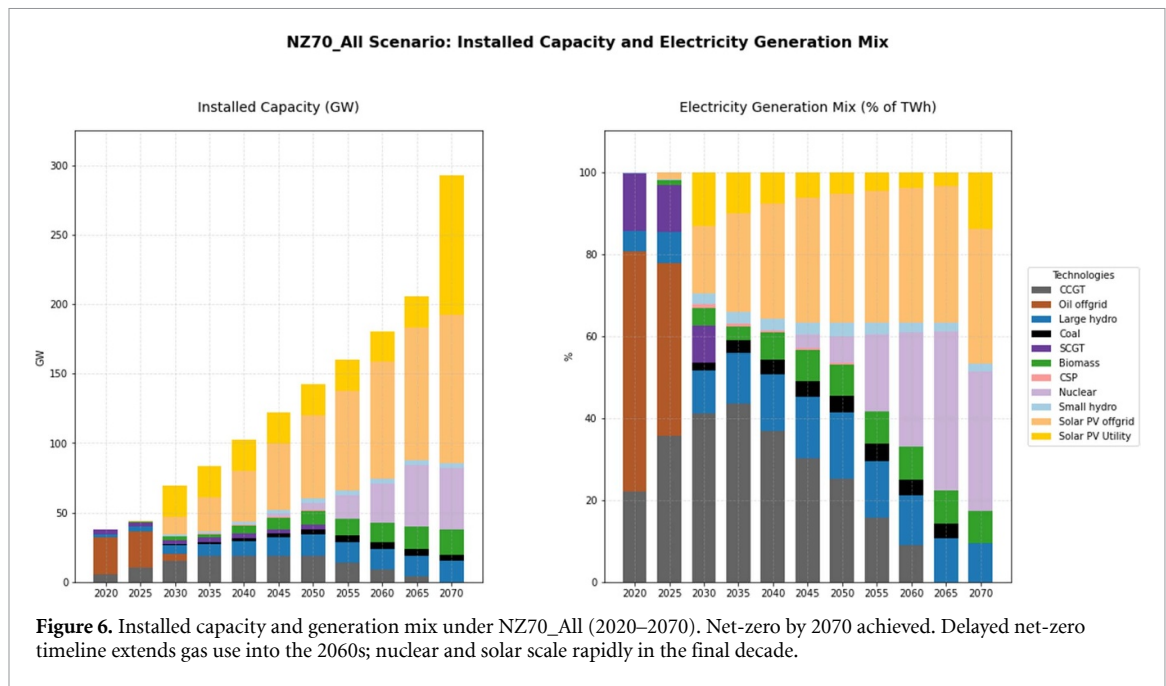
cases), but unlike NZ50_All (which froze at 15), NZ60_All allows modest further gas additions in the 2030s.

The model results also indicate a delayed nuclear ramp up. Nuclear deployment is postponed relative to NZ50_All. By 2050, nuclear power is still relatively low at 8 GW compared to NZ50_All that had 27 GW in same year. The urgency to build nuclear is less because gas and some emissions can continue till 2060. Hydros (small and large) are still maxed out. Large hydro capacity reaches 15 GW by 2050 in NZ60_All as well (since hydro is economic to develop regardless of the exact net-zero date (Ullah *et al* 2024)). Through 2050, utility-scale PV remains modest in the NZ60_All scenario. However, the model again prioritizes nuclear in the 2050s rather than oversize PV. After 2050, as gas is phased out and demand grows, PV picks up: by 2060, utility PV reaches ~51 GW, and by 2070 it is ~72 GW (interestingly the same 72 GW as in NZ50_All at 2070). Wind remains negligible as in other net-zero cases. Interestingly, even by 2060, the NZ60_All scenario still does not build any gas + CCS. Our analysis shows nuclear outcompetes CCS economically, so the model never opts to build gas-CCS when nuclear is an option, even with a 2060 target. The net-zero is achieved by replacing gas with nuclear and renewables directly by 2060.

3.1.4. Net-zero by 2070

Figure 6 presents the results of the NZ70_All scenario while the results of net-zero 2070 scenarios with technology restrictions are presented in the supplementary file (section 1.3). The trends in NZ70 scenarios show an even more relaxed early phase, followed by rapid changes in the 2060s. In the NZ70_All case (figure 6), there is relatively prolonged gas usage. Here, unabated gas (CCGT) capacity continues to grow and persists well into the 2050s. CCGT rises to 15 GW by 2030 (like the 2060 case), and unlike earlier targets, it does not stop there—the model allows some further increase to ~19 GW in 2050. This indicates that under a 2070 target, gas plants can operate up to the late 2060s before being completely retired. No stranded gas asset is observed in 2070 in the NZ70_All. However, as restrictions step in, stranded capacity of CCGT rises to 7.2 GW as observed in NZ70_Restr_Nuc + CCS scenario. Essentially, fossil generation is phased out at the last possible moment in this scenario, giving maximum time for existing assets. No CCS is deployed in NZ70_All either; the model still finds it optimal to just use gas and then drop it by 2070, rather than invest in CCS earlier.

With minimal pressure until the 2060s, nuclear build is very slow initially. By 2040, there is no nuclear in NZ70_All (the model does not bother starting nuclear projects until after 2040, since gas is allowed to handle 2040s growth). By 2050, only ~5 GW of nuclear is built but then ramps up more quickly in the 2050s and especially 2060s as the net-zero year nears: 16.7 GW by 2060, and finally 44.4 GW by 2070. So nuclear still reaches a substantial level, but lower than the ~48 GW seen in



2050/2060 scenarios. This slightly lower peak (44 vs 48 GW) might suggest that by stretching to 2070 and using a bit more gas or other resources, the model did not need to build quite as many reactors.

Solar PV in NZ70_All is the least aggressive early on among net-zero cases. By 2050, utility PV is again held at 22.4 GW (the model still does not push PV beyond the minimal level so long as nuclear and gas are handling things). Off-grid PV is 60 GW by 2050 (like all net-zero scenarios, as that seems tied to electrification rather than emissions as discussed in the BAU results). After 2050, with nuclear coming in slowly and gas still partially present, some PV growth occurs, but substantial PV deployment is deferred until the 2060s. Again, none of wind and CSP are built in NZ70_All. The availability of nuclear (even if delayed) and gas for a long period means wind never enters, and CSP is not needed.

3.1.5. Insights from electricity generation mixes

Across all net-zero scenarios particularly NZ50_All, NZ60_All, and NZ70_All, solar PV emerges as the dominant source of electricity generation post-2040, contributing anywhere from 35%–60% of total generation by 2070 depending on the scenario. This is consistent with its explosive growth in installed capacity seen in the capacity results. For instance, in NZ70_restr_nuc, solar PV accounts for over 50% of electricity generation by 2070, compensating for the absence of nuclear generation and the delayed ramp-up of gas-CCS. This reinforces solar PV's role as the backbone of Nigeria's electricity transition under firm technology constraints.

Firm low-carbon technologies are seen to remain essential in unrestricted scenarios (e.g. NZ50_All, NZ60_All). In these scenarios, technologies like nuclear power contribute significantly to firm generation, particularly between 2040–2070. For example, in NZ50_All, nuclear contributes up to ~30% of total electricity generation by 2050–2070. This suggests that early and unrestricted deployment of firm low-carbon generators improves system diversity and helps smooth the integration of high shares of solar PV and other variable renewables. Across nearly all timelines and technology constraint variants (see supplementary file for results of scenario variants with technology restrictions), biomass and large hydro maintain modest but steady shares of total electricity generation typically between 7%–12%. These technologies act as reliable mid-merit contributors, offering dispatchable capacity without the integration challenges of wind or solar.

In all net-zero pathways, coal and SCGTs are effectively phased out by 2050. Our analysis assumes that new gas investments in Nigeria will be CCGT technology. This outcome is consistent across all net-zero timelines, signaling that coal-fired generation is incompatible with net-zero objectives even in restricted technology variants. The NZ70 scenarios generally retain higher shares of CCGT in the 2030s and 2040s, with generation shares as high as 44% in 2030–2040. However, even in these cases, gas-CCS deployment accelerates after 2050 especially where nuclear is restricted, reaching over 20% of electricity

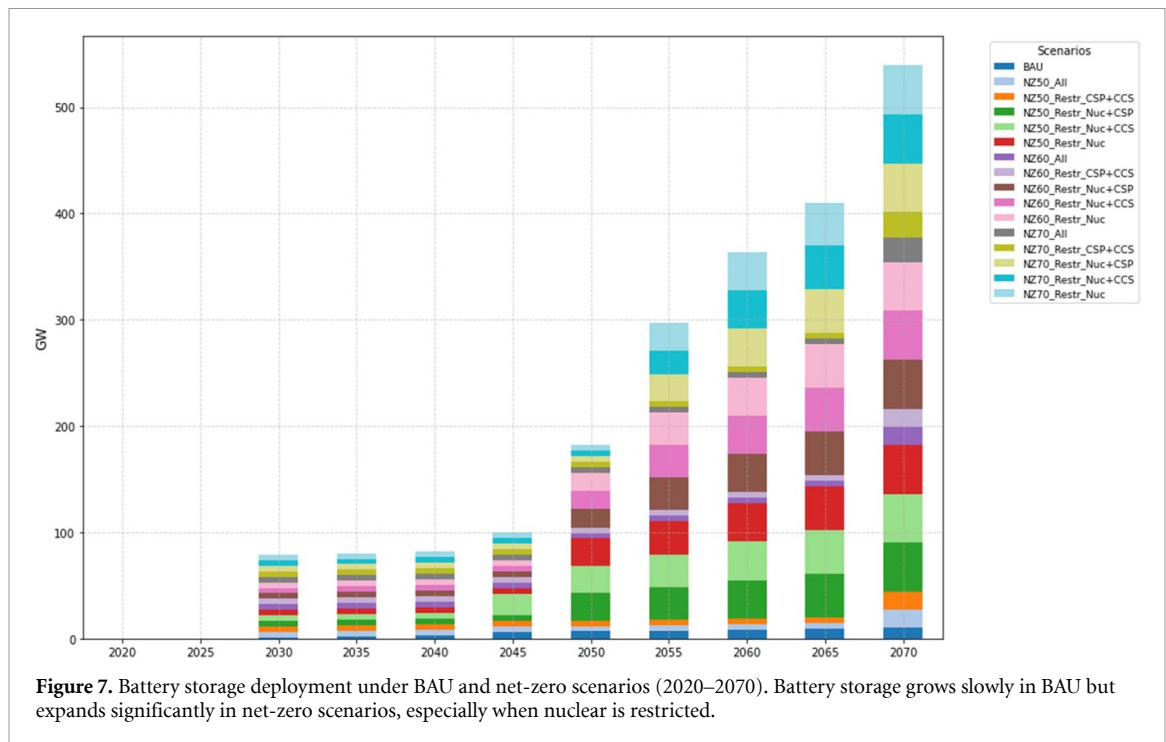


Figure 7. Battery storage deployment under BAU and net-zero scenarios (2020–2070). Battery storage grows slowly in BAU but expands significantly in net-zero scenarios, especially when nuclear is restricted.

generation by 2070. This underlines a core insight: delaying climate action shifts the burden of decarbonization onto more speculative or costly technologies like CCS, which are deployed more aggressively in later decades under NZ70 to catch up with net-zero targets. CSP's role is sensitive to technology restrictions. CSP makes only a small contribution to total generation, especially in the early years; typically under 5% even in some scenarios where it is unrestricted (e.g. NZ50_restr_nuc). This suggests that CSP's competitiveness may be limited without supportive policy frameworks or cost breakthroughs, despite its advantage as a dispatchable renewable source in sunny regions like northern Nigeria.

3.2. Battery energy storage system (BESS) capacity

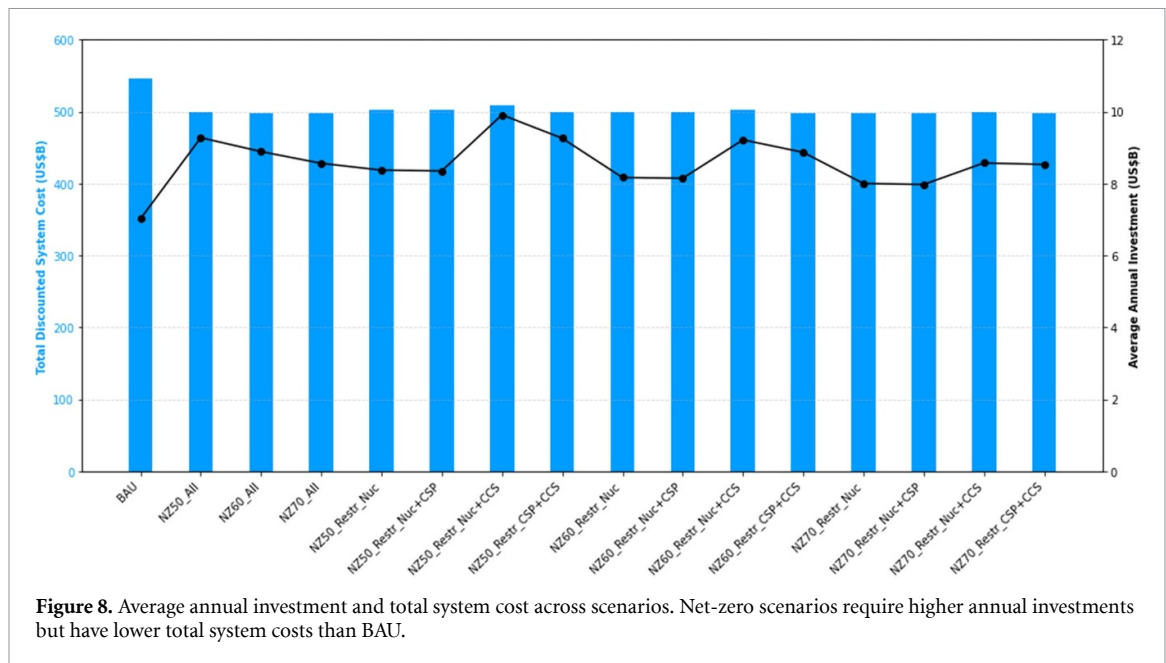
Figure 7 shows the installed capacity of BESS. Our analysis shows that BESS plays a vital role in Nigeria's net-zero electricity transitions, particularly as the penetration of variable renewable energy (mainly solar PV) increases. Across the scenarios analyzed, BESS deployment patterns closely align with both the net-zero timeline and the availability of firm low-carbon technologies. In the BAU scenario, where fossil fuels continue to dominate, battery storage grows slowly and modestly from around 0.7 GW in 2030 to 10.4 GW by 2070, reflecting a system that continues to rely on dispatchable thermal capacity for grid balancing, with limited need for storage.

However, in the net-zero scenarios, we see earlier and steeper growth in BESS capacity, especially when firm low-carbon resources are constrained. In the all-technology scenarios for NZ50, NZ60, and NZ70, battery deployment remains flat at 5.2 GW through the mid-century, before rising modestly to around 17–23 GW by 2070. These scenarios benefit from the presence of nuclear and/or gas w/CCS, which limit the need for deep storage. However, in nuclear-restricted scenarios such as NZ50_Restr_Nuc, NZ60_Restr_Nuc, and NZ70_Restr_Nuc, BESS becomes a central reliability solution. In these cases, storage expands significantly post-2050, reaching 25.7 GW by 2050 and peaking at 46 GW by 2070. This steep increase reflects the need to compensate for the absence of firm baseload power and to firm increasingly high levels of solar PV, which provide more than half of total generation by 2070 in these pathways. The highest BESS deployment consistently appears in the most constrained scenarios, particularly where both nuclear and gas-CCS are unavailable.

3.3. Cost implications

3.3.1. Cost dynamics across net-zero timelines

Figure 8 outlines the average annual investment (generation, storage and T&D) and the total discounted system cost for different scenarios and their corresponding variants within the 2050, 2060, and 2070 net-zero pathways for Nigeria. Our analysis indicates that Nigeria would have to invest substantially in its electricity assets within the modeling horizon (2020–2070). Under the BAU scenario, the total discounted system cost is projected at US\$ 546 billion, while the average annual investment is about US\$ 7 billion.



When comparing the cost implications of the three net-zero timelines, it is evident that the urgency of transition has a strong influence on investment patterns. The NZ50_All scenario, representing a 2050 net-zero target with unrestricted technological deployment, demands the highest average annual investment at US\$ 9.3 billion per year. This elevated investment requirement is consistent with the compressed time frame for decarbonization, necessitating a front-loaded deployment of clean technologies and early retirement of fossil-based infrastructure. In contrast, the NZ60_All and NZ70_All scenarios require average annual investments of US\$ 8.9 billion and US\$ 8.6 billion respectively. The reduced annual investment figures in these cases reflect the extended time horizons, which allow for a more staggered deployment of capital-intensive technologies.

However, the picture shifts when considering the total discounted system cost, which captures the present value of all investments and operational expenditures over the modeled period. The NZ70_All scenario exhibits the lowest total discounted system cost at US\$ 497.3 billion, marginally below the NZ60_All cost of US\$ 498.1 billion and NZ50_All at US\$ 499.8 billion. Interestingly, the BAU scenario, which does not pursue net-zero emissions, incurs the highest system cost of US\$ 546 billion.

3.3.2. Technology restriction trade-offs

Within the NZ50 timeline, imposing restrictions on firm low-carbon technologies has measurable effects on both annual investment and total system cost. The NZ50_Restr_Nuc variant, which excludes nuclear power, reduces the average annual investment to US\$ 8.4 billion, indicating a shift towards less capital-intensive technologies. However, this adjustment leads to an increase in total system cost to US\$ 502.9 billion, suggesting that the substitution of nuclear with variable renewables and storage introduces long-term inefficiencies. Similarly, the NZ50_Restr_Nuc + CSP variant maintains a comparable annual investment of ~US\$ 8.4 billion and a total cost of US\$ 503 billion, demonstrating that CSP's exclusion has a minimal incremental impact when nuclear is already unavailable.

The NZ50_Restr_Nuc + CCS scenario presents the highest cost profile within the NZ50 family of scenarios, with an average annual investment of ~US\$ 10 billion and a total discounted cost of US\$ 508.2 billion. This scenario restricts both nuclear and gas-CCS, forcing the system to rely more heavily on intermittent solar and extensive battery storage. The result is a substantial increase in both capital requirements and total system cost, confirming the critical role of firm generation technologies in cost containment.

In the NZ60 scenario family, similar patterns emerge but with reduced sensitivity to technology restrictions, likely due to the more gradual transition timeline. The NZ60_Restr_Nuc and NZ60_Restr_Nuc + CSP variants both exhibit average annual investments of approximately US\$ 8.2 billion, with identical total system costs of US\$ 499.6 billion. These figures are only marginally above the NZ60_All baseline, indicating that nuclear exclusion within a 2060 timeline has a limited impact on system costs, perhaps due to increased flexibility in substituting other technologies over a longer period. In

contrast, the NZ60_Restr_Nuc + CCS variant again incurs relatively higher costs, with an annual investment of US\$ 9.2 billion and a total system cost of US\$ 502 billion. This outcome aligns with the pattern observed in NZ50 scenarios, where simultaneous restriction of two major firm technologies significantly raises system costs. The NZ70 scenario family shows similar insights with the lowest sensitivity to technology restrictions among the three timelines.

4. Discussion and policy implications

4.1. Electricity system planning implications

All modeled net-zero scenarios see a pivot away from fossil fuels toward renewables, but an earlier target (2050) entails a much more abrupt transition. For example, a 2050 net-zero pathway requires rapidly scaling solar PV (both utility-scale and distributed) to supply roughly 37%–55% of electricity generation by mid-century, along with large investments in firm low-carbon resources (nuclear and gas-CCS) to maintain reliability. This scenario strands a large fraction of existing gas capacity, as supply must be decarbonized quickly. Delaying the net-zero target to 2060 or 2070 allows for a more gradual rollout: continued use of some unabated gas generation through the 2030s, and more time to build out renewable capacity. In practice, our results suggest that later targets reduce the volume of stranded gas assets by nearly a factor of three and lower near-term investment pressures, while still achieving net-zero by the deadline. These findings echo the results of Nana and Dioha (2024), showing that postponing Nigeria's net-zero targets beyond 2050 can leverage lower renewables costs and less negative-emissions technologies reliance.

Allowing a wide portfolio of clean technologies yields the most cost-effective transition, whereas restrictions on certain technologies force less cost-optimal pathways. In scenarios where nuclear power, CCS, or CSP are prohibited, other resources must compensate. For instance, if nuclear is excluded (irrespective of the net-zero target year), the model leans heavily on gas plants equipped with CCS or other dispatchable renewables plus storage to ensure grid stability. Removing both nuclear and CCS leaves CSP as a more important baseload firming resource, but at a higher capital cost, as well as a much larger battery storage requirement. These results align with broader comparative findings that excluding firm low-carbon technologies often leads to higher costs and reliability risks (Baik *et al* 2021). In our Nigerian case study, nuclear (a clean firm power) consistently appears in the optimal mix of all full-technology scenarios, contributing roughly one-third of generation (33%–41% depending on assumptions). This firm output smooths variability and reduces the need for as much battery storage. When nuclear is restricted, costly substitutes step in. Policymakers should therefore carefully weigh the trade-offs of excluding such technologies.

The trade-offs extend to renewables deployment as well. Solar PV (utility and off-grid) dominates all pathways due to Nigeria's exceptional irradiance and projected solar PV cost reduction. However, hydro can contribute balancing power, yet its expansion faces technical potential constraints. Without sufficient renewable alternatives, the system risks intermittency issues. In essence, a balanced approach that includes both high-penetration renewables and firm low-carbon sources yields the lowest overall system costs and highest reliability. Global analysis suggests that restricting key technologies can escalate the costs of the energy transition and endanger energy security thus requiring a portfolio of technologies for net-zero (IEA 2023b); Nigeria's results highlights this, showing lower total system cost when all options compete freely. In policymaking terms, this implies that Nigeria should keep its technology options open and consider incentives (or at least avoid bans) for nuclear and gas-CCS to achieve an affordable transition.

4.2. Investment and cost implications

Our model indicates that achieving net-zero by mid-century would necessitate on the order of US\$ 7–10 billion per year in energy sector investment (figure 8), over ten times above the Nigerian power sector budget of 2025 (Aina 2024). A cross-scenario synthesis of our cost analysis reveals several important insights. First, the average annual investment is strongly correlated with the pace of decarbonization: faster net-zero transition targets such as 2050 require more rapid capital mobilization, while slower pathways like 2070 target allow for more gradual investment profiles. The total system cost in the model is lower under net-zero scenarios than under BAU when lifetime costs (including fuel) are considered, reflecting these savings and the declining costs of renewables. Importantly, allowing a full set of low-carbon technologies yields the lowest-cost pathways. Scenarios restricting nuclear or gas-CCS consistently show higher system costs because they force the system to build more expensive alternatives. Therefore, policymakers should note that investments in diverse clean technologies not only lower emissions but also improve economic outcomes in the long run.

It is also worth noting that delaying the net-zero deadline does not dramatically reduce the overall investment required to decarbonize; it simply shifts costs into the future and somewhat changes their composition. Our scenarios show that the total net-present value of expenditures is comparable whether the target is 2050, 2060, or 2070. Thus, the choice of timeline involves a trade-off between paying costs earlier (with higher short-term rates of build-out and stranded assets) versus paying similar costs later with more gradual change. This finding is congruent with the findings of (Panos *et al.* 2023, Nana and Dioha 2024). Then, the ‘burning question’ is: how will Nigeria—and indeed other sub-Saharan African nations with similar electricity system—rally the resources for an earlier transition?

The cost figures presented here are based on techno-economic modeling and assume optimistic deployments, but real-world factors could inflate costs. Issues like bureaucratic delays, permitting hurdles, and procurement inefficiencies can drive overruns. At the same time, Nigeria’s volatile macroeconomic environment complicates planning. Exchange rate fluctuations—as Nigeria has seen sharp currency losses recently raise the local currency costs of imported solar panels and even nuclear components. Inflation can also erode the value of budget allocations. Given these risks, robust financial planning and risk mitigation are essential (Isah *et al.* 2023).

4.3. Implications for cross-border electricity trade

Cross-border electricity trade offers an important opportunity to improve reliability and reduce costs in Nigeria’s net-zero pathways. West Africa has rich yet spatially varied resources across countries in the sub-region. Integrating Nigeria more fully into the West African Power Pool (WAPP) could allow surplus renewable generation to be exported when available and imports when needed. Indeed, the Economic Community of West African States (ECOWAS) has prioritized projects like four 150 MW regional solar parks and high-voltage interconnections (OMVG and CLSG links) precisely to foster such trade (IRENA Coalition for Action 2023).

Enabling trade could significantly lower system costs. The World Bank estimates that increased regional power exchange in West Africa could save US\$ 5–8 billion per year in the aggregate, by allowing countries to import electricity at lower cost and better match supply and demand profiles (World Bank 2018). For Nigeria, this means less need to over-build domestic capacity and storage to guarantee reliability. For example, Nigeria could export abundant midday solar to neighboring countries and conversely import hydropower from countries like Ghana. It could also provide a market for Nigeria to monetize excess generation, creating a new export revenue stream.

However, realizing these benefits requires both infrastructure and policy alignment. While some interconnectors exist, they must be upgraded and expanded. Legal and commercial frameworks for cross-border trade (power purchase agreements, regional market rules, etc.) need harmonization. At a minimum, Nigeria should actively participate in regional planning forums and bilateral grid studies to ensure its net-zero grid is compatible with neighbors. The trade-enhancing policies identified in the ECOWAS Renewable Energy Policy and implemented through national plans should be supported (e.g. adopting common technical standards and facilitating third-party access to interconnectors) (Maman Ali and Yu 2021).

4.4. Policy recommendations

Our findings suggest several actionable strategies for Nigerian policymakers. First, maintain a diverse technology portfolio. The analysis makes clear that no single technology can satisfy the demand at lowest cost under a net-zero target; rather, a mix of solar PV, hydropower, nuclear, CCS-equipped gas, and storage provides a balanced and resilient system. Policies should therefore encourage—or at least not prohibit—the deployment of each viable clean technology. For example, nuclear energy should remain on the table with a clear regulatory pathway, as it contributes significantly to a least-cost mix in all modeled scenarios where it is unconstrained. Similarly, the government could support pilot CCS projects to preserve optionality.

Second, Nigerian policymakers need to aggressively scale up renewable deployment, especially solar. Given Nigeria’s high solar resource potential (Ogunmodimu and Okoroigwe 2018, IRENA 2023), large-scale PV and distributed solar are the backbone of all transition scenarios analyzed here. Policymakers should streamline permitting and land acquisition processes for solar farms, upgrade transmission corridors to solar-rich northern region of Nigeria. The government might also introduce tariff reforms or auction schemes to attract private investment in renewables (solar PV), as the model shows their dominance is both inevitable and cost-effective especially in scenarios with constrained firm technologies.

The transition will not be resilient without energy storage and grid flexibility. This means not only batteries, but also digital/smart grids and flexible demand management. The battery capacity in all net-zero scenarios (over 46 GW in NZ70_Restr_Nuc scenario) highlights the scale of battery infrastructure

needed. Policies should facilitate the integration of storage by, for example, allowing storage assets to participate in auctions or providing time-of-use pricing. Regulatory updates may also be needed to classify storage separately from generation and allow it to earn revenues for ancillary services (Elshurafa 2020, Rancilio *et al* 2024).

Another policy issue for the government is the need to invest in institutional capacity and good governance. Implementing the electricity transition will require strong, transparent institutions to oversee large infrastructure programs (Edomah 2021). The federal government should bolster the Nigeria Energy Transition Office and agencies like the ECN, Nigeria Electricity Regulatory Commission and the Rural Electrification Agency with technical expertise and clear mandates. Anti-corruption measures and project management best practices will be essential to ensure that high planned investments translate to actual clean energy delivered. Interagency coordination should be improved to align power planning with climate and broader economic development goals (Edomah *et al* 2017, Gungah *et al* 2019).

Nigerian policymakers need to consider mobilizing financing through partnerships. Given the local funding gap, Nigeria should continue to seek international support (grants, concessional loans, guarantees) as well as private capital (Dioha and Kumar 2020). Public-private partnerships should be aggressively pursued in generation, transmission, and distribution projects. Special emphasis could be placed on off-grid solar to leverage private developers, given its potential to rapidly increase electricity access. The government may also consider improving its existing domestic bond issuance (including green bonds) to raise funds (Ngwenya and Simatele 2020). In all cases, clear and stable policy signals (such as predictable tariffs) will be needed to attract investment. Finally, the government needs to plan regionally and inclusively. Coordination with neighboring countries through ECOWAS and WAPP can open up more options (as discussed above). Regional cooperation on energy resources sharing, emergency reserves, and grid code standards will ease integration.

Our analysis is not without some caveats and thus opens the window for further research. Due to computational limitations, the modeling is conducted at the national level using time slices, without incorporating full 8760 hourly data of the year or sub-national data, such as transmission and distribution constraints. Operational constraints, including system inertia and ancillary services markets, are not explicitly considered and the entire analysis is broadly limited to lack of Nigeria-specific data. While acknowledging these limitations, the study provides valuable insights into Nigeria's pathway to a net-zero emissions electricity system and is relevant for other developing regions.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Supplementary Information available at <https://doi.org/10.1088/2516-1083/ae2bd7/data1>.

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Declaration of AI-assisted technologies in the writing process

The python codes used for the K-means cluster analysis and figures were cleaned using AI. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

ORCID iDs

Michael O Dioha  0000-0001-6983-6752

Norbert Edomah  0000-0003-2893-4406

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